Neuroscience Research 215 (2025) 37–46

Contents lists available at ScienceDirect

Neuroscience Research

journal homepage: www.sciencedirect.com/journal/neuroscience-research

Review article

Inference of monosynaptic connections from parallel spike trains: A review

Ryota Kobayashi [a,b,*], Shigeru Shinomoto [c,d]

[a] Graduate School of Frontier Sciences, The University of Tokyo, Chiba 277-8561, Japan
[b] Mathematics and Informatics Center, The University of Tokyo, Tokyo 113-8656, Japan
[c] Graduate School of Biostudies, Kyoto University, Kyoto 606-8501, Japan
[d] Research Organization of Open Innovation and Collaboration, Ritsumeikan University, Osaka 567-8570, Japan

ARTICLE INFO

Keywords:
Spike trains
Monosynaptic connection
Cross-correlation
Transfer entropy
Generalized linear model

ABSTRACT

This article presents a mini-review about the progress in inferring monosynaptic connections from spike trains of multiple neurons over the past twenty years. First, we explain a variety of meanings of "neuronal connectivity" in different research areas of neuroscience, such as structural connectivity, monosynaptic connectivity, and functional connectivity. Among these, we focus on the methods used to infer the monosynaptic connectivity from spike data. We then summarize the inference methods based on two main approaches, i.e., correlation-based and model-based approaches. Finally, we describe available source codes for connectivity inference and future challenges. Although inference will never be perfect, the accuracy of identifying the monosynaptic connections has improved dramatically in recent years due to continuous efforts.

1. Introduction

It has been about a hundred years since it was discovered that neurons generate electrical pulses and that these electrical pulses carry information between neurons. As it became possible to chronically measure the activity of a neuron during animal behavior, scientists began to investigate how individual neurons are involved in a neural response to stimuli. In recent years, it has become possible to record spike signals from multiple neurons in parallel, and the number of such neurons has increased over the years, now exceeding a thousand (Stevenson and Kording, 2011; Steinmetz et al., 2021), and even more than ten thousand with optical recording technologies (Ota et al., 2021; Kim and Schnitzer, 2022). Although this number is still small compared to the total number of neurons in the brain, it is likely to increase in the future, and this information will eventually yield a great deal of knowledge about neural information processing.

A straightforward way to study the function of the animal's nervous system is to examine how stimuli are represented in neural activity in the brain and how this neural activity regulates the animal's behavior. For example, neuronal representation in the visual system has been analyzed by measuring individual cells in the visual cortex and exploring the visual stimuli to which these cells respond (Hubel and Wiesel, 1977; Fujita et al., 1992); pioneering studies of the motor control of animal behavior have revealed the role of individual neurons in operantly conditioned movements by recording neuronal activity in the sensorimotor cortex of animals during behavior (Evarts, 1968). These early studies considered a single neuron as representative of the myriad of cells in each brain region that work for a single function. Recent advances in measurement technology have made it possible to record the activity of many individual cells in parallel and to elucidate the role of multiple neurons in animal behavior. Assuming independence among neurons, information about the stimulus-response relationship of individual cells increases in proportion to the number of cells measured simultaneously.

In addition, recent measurement technology not only allows us to obtain information about the correlation between stimulus and single cell activity, but also provides information about the correlation of activity between neurons, allowing us to capture information transmission at the level of individual neurons. The amount of information about correlations between neurons increases in proportion to the square of the number of cells measured, and it increases faster than the amount of information about the stimulus-response relationship. In 1967, more than half a century ago, when it was barely possible to measure the activity of multiple neurons simultaneously, Perkel, Gerstein, and Moore proposed a method for inferring monosynaptic connections between neurons by evaluating the interdependence of multiple neuronal spike trains (Perkel et al., 1967).

However, neurons in the brain are densely interconnected and

* Corresponding author at: Graduate School of Frontier Sciences, The University of Tokyo, Chiba 277-8561, Japan.
E-mail address: r-koba@k.u-tokyo.ac.jp (R. Kobayashi).

https://doi.org/10.1016/j.neures.2024.07.006
Received 19 March 2024; Received in revised form 12 July 2024; Accepted 19 July 2024
Available online 2 August 2024
0168-0102/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

influence each other in complex ways. There is often interdependence in the firing of neurons even when they are not directly connected, and this classical method has been shown to make many false inferences. In this article, we review the advances in the methods for inferring monosynaptic connections from spike data over the last twenty years. Finally, we discuss the limitations and future challenges of current methods.

2. Variety of meanings for "neuronal connectivity"

The nervous system in the brain consists of a myriad of neurons that transmit signals through synaptic connections to produce the neural response of an animal. Bottom-up research on brain function analyzes the circuit structure of how neurons are connected via synapses and elucidates the mechanisms by which these connections are plastically modified. Before getting to the main topic, however, we must first mention that the term "connectivity" or "connection" is used in three ways depending on the field of neuroscience, i.e., structural connectivity, monosynaptic connectivity, and functional connectivity.

2.1. Structural connectivity

Structural connectivity means how neurons are anatomically connected. Studies that attempt to analyze the structure of neural circuits include efforts to reconstruct neurons in three-dimensional space using electron microscopy (Helmstaedter et al., 2013; Scheffer et al., 2020). To study the structural connections between distant neurons or between brain regions, connections are first visualized at the neuronal level using horseradish peroxidase and other techniques (Binzegger et al., 2004). Structural connectivity at the macroscopic level between brain regions is analyzed using diffusion tensor imaging (DTI) that visualizes the bundles of nerve fibers. The structural connections obtained in this way have been used to simulate neural circuits to discuss the function of columns and the impact of a single neuronal spike on the dynamics at the level of the whole brain (Izhikevich and Edelman, 2008; Schmidt et al., 2018).

2.2. Monosynaptic connectivity

The mechanism of signal propagation between neurons is that action potentials (spikes) generated in one neuronal cell (presynaptic neuron) propagate through axons and, upon arrival at synapses at the axon terminals, chemical transmitters are released, causing changes in the membrane potential of a postsynaptic neuron (Fig. 1a, b). When the membrane potential of a neuron reaches the threshold, the neuron generates a spike. Monosynaptic connectivity is defined as a direct influence between neurons in their membrane potential (i.e., EPSP or IPSP). To confirm the monosynaptic connections, we can stimulate one neuron to generate an action potential and observe whether the activation of the neuron results in a change in the membrane potential of another neuron (Fig. 1c, d). If this measurement can be done, not only the presence or absence of synaptic connections, but also the strength of the connections can be measured in units of excitatory/inhibitory postsynaptic potentials (EPSP/IPSP) (Thomson et al., 2002; Song et al., 2005).

Although synaptic connections can be rigorously determined by intracellular recording, intracellular recording in behaving animals is challenging, and the number of cells that can be measured simultaneously is limited to a few, even for brain slices (*in vitro*). In contrast, technological innovations are continuously increasing the number of cells that can be measured simultaneously using extracellular recording.

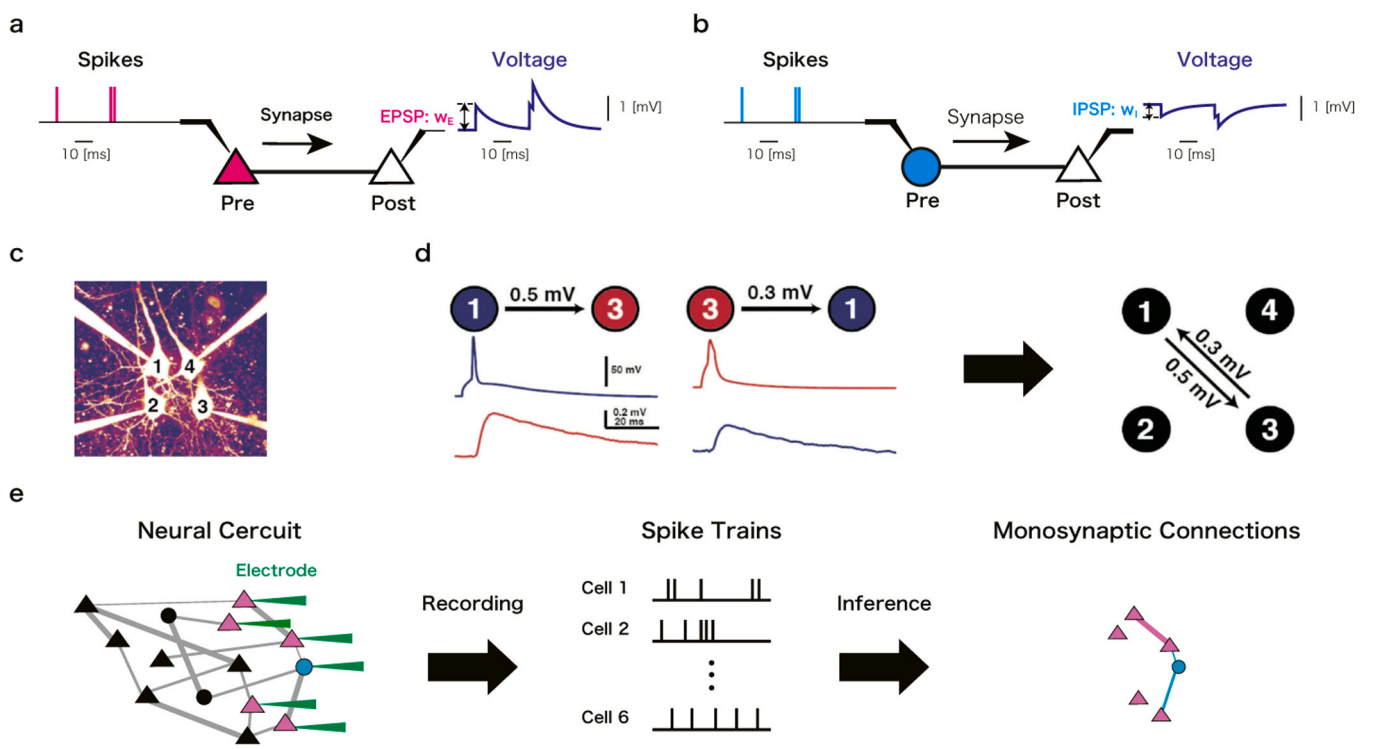

Fig. 1. Monosynaptic connections between neurons. (a) and (b): Signal transmission between neurons. Due to the synaptic connection, the membrane potential of a postsynaptic neuron (Post) increases (a) or decreases (b) after an action potential of a presynaptic neuron (Pre). Whether the membrane potential increases or decreases depends on the type of presynaptic neuron. While an excitatory neuron (triangle) increases the membrane potential of the postsynaptic neuron, an inhibitory neuron (circle) decreases it. The strength of the monosynaptic connection is defined as the change in potential (EPSP or IPSP). (c) and (d): Experimental measurement of the monosynaptic connections. Using quadruple whole-cell recordings (c), the EPSP or IPSP can be measured by evoking action potentials in each of the four neurons and measuring the change in the voltage in the other neurons (d). Panels (c) and (d) are adapted and modified with permission from Song et al. (2005). (e): Inference of monosynaptic connections from spike trains of multiple neurons. The monosynaptic connections can be inferred from the correlation between spike trains. In this review, we focus on the advances in the method to infer/estimate the monosynaptic connections from spike data.

The presence of a monosynaptic connection can be inferred from the correlation between spike trains with a time difference of a few milliseconds (Fig. 1e). Although the inference is subject to some error, it may be possible to improve the accuracy of the inference by improving the data analysis method. In this paper, we review the methods that have been developed for this purpose.

2.3. Functional connectivity

Neurons in the brain are connected at multiple levels, even if they are not connected monosynaptically, and their neural activity may not be independent of each other. Many studies have attempted to detect correlations of activity between different brain regions, particularly in humans, using non-invasive measurements such as fMRI, in search of potential features associated with neurological and psychiatric disorders (Lynall et al., 2010; van den Heuvel and Sporns, 2019). They refer to the correlation as "functional connectivity", but this simply means an apparent correlation between brain regions or between neurons, and it "does not provide direct insight into how these correlations are mediated" (Friston, 1994). It should be noted that the existence of a strong functional connection (direct or indirect) between two given elements cannot be attributed to the physical connection between the two elements.

3. Methods for inferring monosynaptic connections

This review focuses on studies that aim to infer or estimate direct monosynaptic connections between neurons. We will refer to the studies we focus on as "inferring monosynaptic connections" to distinguish them from studies that simply detect activity correlations as "functional connections".

The number of papers aiming at the inference of monosynaptic connectivity has increased in recent years (Fig. 2). We searched Web of Science for papers containing the keywords "connectivity, inference, spike," "connectivity, estimation, spike," or "connectivity, reconstruction, spike." in the title or abstract. The search returned 308 articles. We found 124 articles that addressed the problem of inferring monosynaptic connections between neurons. Even excluding the 8 articles that used intracellular recording to measure synaptic connectivity and one article that focused on a graphical user interface (GUI) to analyze recorded data, we still have 115 articles.

Methods for inferring the monosynaptic connections between neurons can be divided into two categories: 1) methods based on the statistical characteristics of the spike correlation between neurons and 2) methods based on the model of the dynamic spike generation process of the recorded cells. In this paper, we call them "correlation-based" and "model-based", respectively (Table 1). The former has sometimes been called "model-free" (Stetter et al., 2012; Vicente et al., 2011; de Abril et al., 2018). Recently, however, it has been realized that applying a statistical model such as the generalized linear model to the spike correlation is an efficient way to analyze the data (e.g. Kobayashi et al. (2019))), and we have changed the category name accordingly. In addition, information-theoretic quantities, such as mutual information, are measures of correlation, because they evaluate the dependence between two random variables (Cover and Thomas, 1991). In the following sections, we will further subdivide them with brief methodological explanations, and finally introduce open source codes that are currently available.

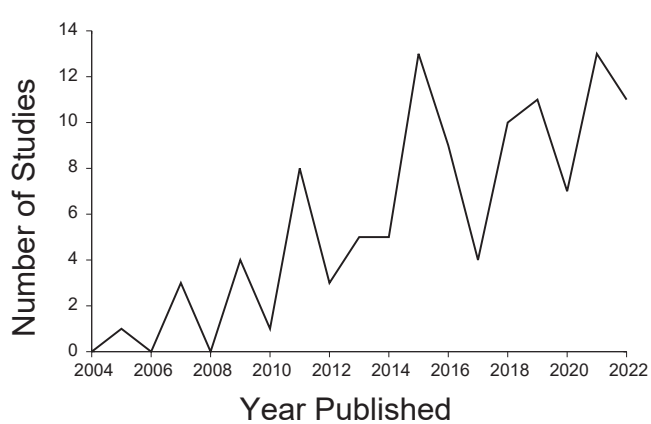

Fig. 2. Number of papers that contain keywords of "connectivity, estimation, spike," "connectivity, inference, spike," or "connectivity, reconstruction, spike.".

3.1. Correlation-based approach

Methods based on the spike-occurrence correlation of two cells occupy 53 % of all studies of monosynaptic connection estimation. Here we further subdivide them by the terms "cross-correlation," "information theory," and "others" (Table 1). The main feature of the correlation-based approach is its ease of use compared to the model-based approach. The classical cross-correlation method (Perkel et al., 1967) is easy to interpret the inference result and simple enough to implement manually. In addition, the source codes of advanced methods are available in MATLAB, Python, or web applications (Table 2).

3.1.1. Cross-correlation method

The membrane potential of a neuron fluctuates due to the input from other neurons transmitted via synaptic connections. A spike arriving at an excitatory/inhibitory synapse increases/decreases the membrane potential of a target neuron, i.e., the postsynaptic neuron (Fig. 1a, b). Accordingly, the probability of a spike occurring at the postsynaptic neuron (Post in Fig. 1a, b) increases/decreases for a few milliseconds after the occurrence of a spike at another neuron (Pre in Fig. 1a, b).

The classical cross-correlation (classical CC) method attempts to find the existence of synaptic connections between neurons by analyzing the cross-correlation of spike occurrences of two neurons to detect the causal influence due to the synaptic connections (Perkel et al., 1967; Moore et al., 1970). The method consists of the following three steps (Fig. 3a).

1. Obtain a cross-correlation by computing the time of spikes of a target neuron (cell B) measured relative to each spike of the reference neuron (cell A) (Fig. 3a: center).
2. Construct the cross-correlation histogram or cross-correlogram (CCG) (Fig. 3a: right).
3. Determine that there may be a synaptic connection from A to B (or B to A) if the CCG has a statistically significant peak or dip in the region $s > 0$ (or $s < 0$).

There are several variants of the classical method: the spike coherence (Brillinger et al., 1976), the joint peri stimulus time histogram (JPSTH) (Gerstein and Perkel, 1969; Aertsen et al., 1989), and the covariance (Pernice and Rotter, 2013).

The classical CC method has the advantage of high interpretability because the influence of the synaptic connection is visible from the CCG. However, the CCGs often exhibit large fluctuations, which may be

Table 1
Approaches to infer monosynaptic connections.

Category	Method	Papers (%)
Correlation-based	Cross-correlation	25 (22 %)
	Information theory	11 (10 %)
	Others	25 (22 %)
Model-based	GLM	36 (31 %)
	Others	18 (16 %)

Table 2
Available codes for connectivity inference from spike trains.

Method	Language	URL
TSPE (De Blasi et al., 2019)	MATLAB	https://github.com/biomemsLAB/TSPE
GLMCC (Kobayashi et al., 2019)	Python	https://github.com/NII-Kobayashi/GLMCC
	Web application	https://s-shinomoto.com/CONNECT
ExGLM (Ren et al., 2020)	MATLAB	https://github.com/NaixinRen/extended-GLM-for-synapse-detection
Transfer entropy (Ito et al., 2011)	MATLAB	https://code.google.com/archive/p/transfer-entropy-toolbox
CoTETE (Shorten et al., 2021)	Julia / Python	https://github.com/dpshorten/CoTETE.jl
CoNNECT (Endo et al., 2021)	Python	https://github.com/shigerushinomoto/CoNNECT
	Web application	https://s-shinomoto.com/CONNECT
GLM (Pillow et al., 2008)	MATLAB	https://github.com/pillowlab/GLMspiketools

caused by a large number of other neurons that are not recorded, and such large fluctuations may induce many false detections. To overcome this difficulty, there have been several approaches to improve the estimation. In the following, we introduce four main approaches to extend the classical CC method.

- **Shift predictor** (Fig. 3**b**): A CCG constructed from two neurons often exhibits a broad peak near the zero time lag, even when the neurons are not monosynaptically connected. A shift predictor has been proposed to avoid the false inference caused by this peak (Perkel et al., 1967; Toyama et al., 1981; Palm et al., 1988). When an animal is repeatedly exposed to the same stimulus, this activation causes a broad peak in the CCG. The correlation due to the stimulus may be extracted by constructing a CCG between a reference cell (Cell A) and an artificial spike train (Cell B shift) obtained by shifting the response of a trial, which is called the shift predictor. The influence of a monosynaptic connection is expected to appear in the difference between the original CCG and the shift predictor.
- **Jittering** (Fig. 3**c**): While a CCG may also exhibit slow fluctuations caused by background activity, the monosynaptic effect is expected to show up as a sharp peak or dip typically in a few milliseconds near the origin ($s = 0$). The sharp peak or dip near the origin may be destroyed if the original spike times are jittered for several milliseconds, while the slow fluctuations remain. The jittering method estimates the statistical significance of a peak or a dip in the original CCG by constructing many CCGs from jittered spike trains (Fujisawa et al., 2008; Amarasingham et al., 2012).
- **Decomposition** (Fig. 3**d**): This approach extracts the effect of the monosynaptic connections by decomposing the CCG into multiple components. Kobayashi et al. (2019) applied the generalized linear model, called GLMCC, to decompose the CCG into monosynaptic influences and background fluctuations caused by other neurons. Here they assumed that the inputs from many unobserved neurons produce a slow undulation in the CCG, and attempted to absorb the influence by fitting a smooth function, while separately detecting a peak or dip of a few milliseconds caused by monosynaptic impact. GLMCC was able to estimate synaptic connections more accurately than the classical CC and Jittering when applied to synthetic spike trains generated by simulating a network of 1000 Hodgkin-Huxley model neurons and a network of 10,000 leaky integrate-and-fire neurons. Ren et al. (2020) employed a similar model and showed that the estimation accuracy can be improved by taking the neuron type into account. Spivak et al. (2022) proposed to deconvolute the CCG into the effect of spiking of a reference neuron, a target neuron, and other neurons. More recently, Tsubo and Shinomoto (2024) pointed out that a cusp often appears at the origin of many biological CCGs, and proposed a new method, ShinGLMCC, to avoid making false inferences caused by the cusp.
- **Post-processing:** CCGs of *in vivo* data are often jagged, burying a clear peak or dip. There are several suggestions for post-processing to find a peak/dip in a CCG. Pastore et al. (2018) suggested subtracting the filtered mean from the original CCG. De Blasi et al. (2019) used an edge filter used in image processing and processed the CCG. They generated synthetic data by simulating a network of 1000 Izhikevich model neurons and showed that their method provided better estimation.

Finally, we discuss the features and limitations of four approaches: shift predictor, jittering, decomposition, and post-processing. While the shift predictor can efficiently remove the estimation bias caused by external stimuli such as visual inputs, it requires that each pair of neurons can be experimentally stimulated. The other approaches do not require such detailed individual operations. The jittering method has succeeded in removing the false positive connections by increasing the significance level, but it is too conservative and misses many connections that may be involved in information processing (Kobayashi et al., 2019). The decomposition and post-processing approaches were developed to improve inference performance by balancing the conflicting demands of reducing false positives and false negatives.

Application to experimental data

The classical CC method and its extensions are not only validated with synthetic data, but also applied to many experimental data.

- **Classical Cross-Correlation:** Barthó et al. (2004) used the classical CC method to estimate synaptic connections between neurons in layer 5 of the somatosensory cortex and the prefrontal cortex of a rat. The number of connections detected was 0.2 percent of the 60,000 pairs analyzed. The estimation accuracy of the classical CC method was evaluated using synthetic data obtained from a network of 60 Izhikevich neurons (Garofalo et al., 2009) or a network of 5000 Hodgkin-Huxley neurons (Kobayashi and Kitano, 2013).
- **Shift predictor:** Shift predictor has been used to analyze many experimental studies: Cat visual cortex (Toyama et al., 1981), Cat thalamus to visual cortex (Reid and Alonso, 1995), Rat visual cortex (Yoshimura et al., 2005), Rat thalamus to somatosensory cortex (Liew et al., 2021), Monkey V1 and V2 area (Nowak et al., 1999), and Monkey prefrontal cortex (Funahashi and Inoue, 2000).
- **Jittering:** Jittering has been used to analyze rat medial prefrontal cortex (Fujisawa et al., 2008), rat hippocampus (Mizuseki et al., 2009), and rat prefrontal cortex (Schwindel et al., 2014).
- **Decomposition:** Kobayashi et al. (2019) proposed the GLMCC and applied it to hippocampal CA1 and EC data (Mizuseki et al., 2013) measured from a freely behaving rat. The excitatory and inhibitory characteristics of individual neurons determined by GLMCC showed good agreement with those determined by experts based on spike waveforms and CCGs. Furthermore, the cell-to-cell binding probabilities were consistent with those previously identified in slice experiments. Ren et al. (2020) proposed ExGLM and applied it to data measured from cultured cells *in vitro* using a large multielectrode array (512 electrodes). The presumed excitable characteristics were consistent with previous experimental results. Spivak et al. (2022) applied their proposed method to mouse hippocampal CA1 data measured from three freely-moving mice recorded with high-density silicon probes. The results show that the proposed method can remove the effect of burst firing of presynaptic neurons.
- **Post-processing:** Pastore et al. (2018) used their method, FNCCH, to estimate synaptic connections *in vitro* cell culture (4096 units). The results suggest that the *in vitro* network has small-world, scale-free, and rich-club (strongly connected) properties.

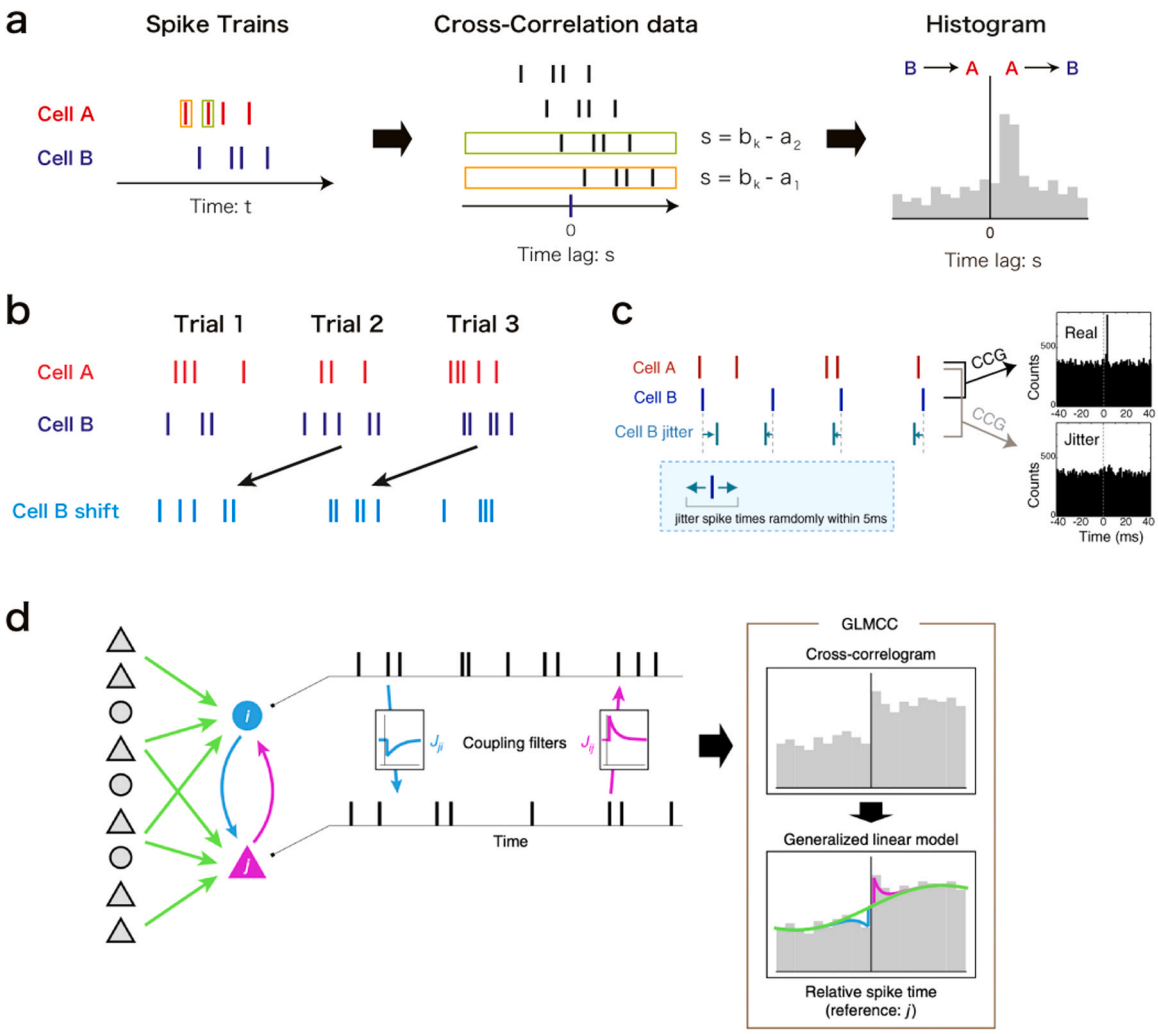

Fig. 3. Classical cross-correlation method and its extensions. (a): Classical cross-correlation method. First, the cross-correlation data is obtained by computing the time lag $s = b_k - a_j$ relative to the reference cell (cell A). Note that a_k (b_k) is the k th spike time of cell A (B). A histogram of the CC data (CCG) is then constructed. If the CCG has a peak or dip in the range of time difference $s > 0$ ($s < 0$), it suggests a synaptic connection from cell A to B (cell B to A). (b): Shift predictor. First, an artificial spike train (Cell B shift: cyan) is generated by shifting the trial of the target cell (Cell B: blue). The shift predictor is then obtained by calculating the CCG between Cell A (red) and Cell B shift (cyan). Finally, the connection is inferred from the histogram obtained by subtracting the shift predictor from the original CCG between cell A (red) and cell B (blue). (c): Jittering. A jittered spike train (Cell B jitter: cyan) is generated by randomly (± 5 [ms]) shifting the spike of the target cell (Cell B: blue). Monosynaptic connectivity is inferred by comparing the CCG between Cell A (red) and Cell B (blue) with the CCG between Cell A (red) and Cell B jitter (cyan). (d): Decomposition. This approach infers the synaptic connectivity by decomposing the CCG into multiple components. For example, GLMCC estimates synaptic connections by decomposing the CCG into the effects from cells other than the two cells, including unobserved neurons (green) and the effects due to synaptic connections between the two cells (cyan, magenta).
(a) Panel (c) is adapted with permission from Fujisawa et al. (2008). (b) Panel (d) is adapted with permission from Kobayashi et al. (2019).

3.1.2. *Information-theoretic measure*

Information-theoretic measures such as joint entropy (Garofalo et al., 2009) or directed information (Quinn et al., 2011; So et al., 2012; Cai et al., 2017) have been used to estimate synaptic connections. Here we introduce the transfer entropy (TE) (Schreiber, 2000), which is the most common among the 11 articles in Table 1, see Vicente et al. (2011) for a review.

TE from one neuron j to another neuron i quantifies the degree to which the spike data of neuron j can improve the prediction of the spike data of neuron i. If there is a strong excitatory synapse from neuron j to neuron i, spike occurrence in neuron j should have a strong impact on spike generation in neuron i, and accordingly the TE should take a large value. In contrast, if there is no synapse from neuron j to neuron i, the activity of neuron j should not affect the prediction of spike data of neuron i, and the TE is expected to vanish (TE = 0).

When computing the TE, spike trains are discretized with a time bin Δ, i.e. it is represented as a discrete time binary series, where $I_k = 1$ ($J_k = 1$) if the i th (j th) neuron generates spikes in the k th time bin, otherwise $I_k = 0$ ($J_k = 0$). The TE from one neuron j to another neuron i is calculated as

$$\mathrm{TE}_{ji} = \sum_{I_k, J_{k-1}, J_{k-1}} p(I_k, I_{k-1}, J_{k-1}) \log \frac{p(I_k | I_{k-1}, J_{k-1})}{p(I_k | I_{k-1})}, \tag{1}$$

where $p(I_k|I_{k-1}, J_{k-1})$ and $p(I_k|I_{k-1})$ represent the conditional probabilities. The advantage of the TE over the classical CC method is that it estimates synaptic connections taking into account the past activity of neuron i itself, as can be seen from the equation (1). This is expected to allow TE to estimate synaptic connections by incorporating neuronal properties such as refractory periods and bursts.

Improvements to the TE have also been proposed. For instance, Ito et al. (2011) proposed Higher Order TE (HOTE), which considers spiking data before d steps (I_{k-1}, I_{k-2}, $\cdots$ I_{k-i}; J_{k-d}, J_{k-d-1}, $\cdots$ J_{k-d-m}) and showed it can improve the accuracy of estimation. Furthermore, the

estimated TE value depends critically on the time bin Δ, which potentially affects the inferred connectivity. For this reason, it has been proposed to fit GLM (Quinn et al., 2011; So et al., 2012) or to estimate the TE directly from interspike interval data (Shorten et al., 2021).

Information-theoretic measures have also been applied to experimental data. For example, the directed information has been used to estimate synaptic connectivity in monkey primary motor cortex (Quinn et al., 2011; So et al., 2012). The extended TE has been used to estimate synaptic connectivity in slice cultures of rodent somatosensory cortex (Ito et al., 2011). More recently, the TE has been used to examine changes in connections between neurons in rat embryo dissociated hippocampal neuron cultures *in vitro* at 6–35 days post-culture to determine how neural networks evolve as a result of maturation (Antonello et al., 2022).

3.1.3. Other methods

This section introduces correlation-based methods that are not based on the cross-correlation method or the information-theoretic measure.

- **Granger causality:** Granger causality (Granger, 1969; Seth, 2007) has also been used to estimate monosynaptic connectivity. Since the original Granger causality analysis is proposed for time series, it needs to be modified for the analysis of point event data (point process) when considering neuronal spike data, e.g., by using the Fourier transform (Nedungadi et al., 2009), introducing a likelihood ratio test for the point process model (Kim et al., 2011), or simply smoothing the spike data (Shao et al., 2015).
- **Spike triggered non-negative matrix factorization (STNMF):** A method combining spike-triggered average and non-negative matrix factorization (STNMF) has been proposed to estimate the synaptic connectivity in the retina (Liu et al., 2017).
- **Others:** Other methods have also been proposed using the spike train metric (Kuroda et al., 2011) and the copulas (Sacerdote et al., 2012). An interesting approach is to apply the artificial neural network to infer the synaptic connectivity. For instance, Endo et al. (2021)) developed a convolutional neural network, CoNNECT, to analyze data recorded from the prefrontal, IT, and V1 cortices of monkeys. More recently, Donner et al. (2024)) developed an ensemble artificial neural network, eANN, to improve connectivity inference.

3.2. Model-based approach

While the correlation-based approach directly analyzes the interdependence of two given spike trains, the model-based approach constructs a mathematical model in which spiking neurons interact. The model-based approach can take account of all recorded neurons simultaneously. The synaptic connections are determined by fitting a large number of model parameters to given spike trains. An advantage of the model-based approach is that it can incorporate time-dependent external influences into the statistical model, including the activity of other recorded neurons and the sensory stimulus. However, the model-based approach is not easy to implement because it requires incorporating the detailed experimental conditions into the mathematical model.

3.2.1. Generalized linear model

Many studies have used the generalized linear model (GLM) (Paninski, 2004; Okatan et al., 2005; Truccolo et al., 2005; Chen and Brown, 2022), see Stevenson et al. (2008b) for a review. The firing rate $\lambda_i(t)$ of the i th neuron is given as a function of an input

$$\lambda_i(t) = F(\text{Input}), \tag{2}$$

where the input is decomposed as

$$\text{Input} = [\text{Base}] + [\text{Stimulus}] + [\text{Refractory}] + [\text{Coupling}]. \tag{3}$$

The GLM (Eq. (3)) assumes that neuronal firing is influenced by the following four factors. The first term [Base] represents the spontaneous firing rate. The second term [Stimulus] represents the effect of the stimulus and is often modeled by the convolution of the external stimulus and the temporal/spatial filter. The third term [Refractory] represents the property of neurons that are less likely to fire immediately after a spike, which is related to the refractory period or spike frequency adaptation (Koch, 2004; Kobayashi and Kitano, 2016). The fourth term [Coupling] represents the monosynaptic effect from other neurons recorded in parallel, and the synaptic connections are inferred from the parameters in this term. The coupling effect is modeled by the sum of filters that describe the effect of a spike from another recorded neuron on the i-th neuron. Several coupling filters have been used in previous studies. For example, the exponential function was used to infer synaptic connectivity from simulated data (Kobayashi and Kitano, 2013; Zaytsev et al., 2015), and the sum of spline functions (the window of 5 ∼ 100 [ms]) was used to infer the physiological circuit in the crab (Gerhard et al., 2013).

One of the reasons that GLM is often used is the applicability of the maximum likelihood method. The maximum likelihood method has been shown to provide more accurate parameter estimation than the method of moments (Brillinger, 1975) or other methods (Berry and Meister, 1998; Chornoboy et al., 1988; Paninski, 2004). For GLM, it has also been shown that the maximum likelihood method does not have a local maximum (Paninski, 2004). The input-output functions F can be the exponential function (Jolivet et al., 2006; Kobayashi and Shinomoto, 2007) or the Relu function (Ermentrout, 1998; Kobayashi, 2009). GLMs whose input-output function is a Relu function are also called Hawkes process (Hawkes, 1971), for which fast estimation algorithms have been developed (Lambert et al., 2018).

It should be noted that the number of parameters in the GLM tended to be too large and overfitting occurred when applied to experimental data (Stevenson et al., 2008b). Research has been conducted to overcome this problem. One approach is to use L1 (Lasso) regularization to reduce the effective number of parameters (Paninski, 2004; Zaytsev et al., 2015; Lambert et al., 2018). For example, Pillow et al. (2008) estimated significant functional connections between retinal ganglion cells by incorporating the physiological knowledge that synaptic connections are sparse using the Group Lasso method (Yuan and Lin, 2006). Another approach is to reduce the effective number of parameters by Bayesian estimation (Gerwinn et al., 2010). For example, Stevenson et al. (2008a) developed an estimation method that incorporates the smoothness of the postsynaptic potential and the sparsity of the connections into the prior distribution. Chen et al. (2011) reported that Hierarchical Bayesian modeling is more accurate than L1 regularization in estimating connections.

Applications to synthetic and experimental data

Below we summarize the applications of GLMs to synthetic and experimental data.

- **Applications to synthetic data:** The validity of GLMs can be tested by estimating synaptic connectivity from spike data obtained by numerical simulation of an interconnected network of neurons. However, most GLM studies have estimated functional connectivity, and few studies have examined the correspondence between functional connectivity and actual synaptic connectivity (exceptions are Kobayashi and Kitano (2013) and Zaytsev et al. (2015). Zaytsev et al. (2015) proposed a fast algorithm for estimating synaptic connections, and the proposed method was applied to simulated data based on a network of 1000 spiking model neurons. It has been shown that the proposed method can accurately and quickly estimate synaptic connections even when 1000 cells are measured simultaneously.
- **Applications to biological data:** An example of experimental validation of GLM is a study of the stomatogastric ganglion (STG)

circuit in the crab (*Cancer borealis*) (Gerhard et al., 2013). In this study, the synaptic connections between neurons were estimated from spike data measured from the STG. The results showed that connections estimated using GLM were consistent with known physiological connections. In contrast, connections estimated using Granger causality and firing rate were not consistent with physiological connections.

- GLM has also been applied to the retina (Pillow et al., 2008; Gerwinn et al., 2010), hippocampus (Okatan et al., 2005; Xia et al., 2019), and motor system (Stevenson et al., 2008a). The neuroscientific significance of the inferred connections has also been investigated. For example, Pillow et al. (2008) showed that inferred connections between two types of cells (ON and OFF retinal ganglion cells) in the monkey retina improved the accuracy of predicting firing rates and decoding visual stimuli. This result suggests that the inferred connections may contribute to visual information processing in the retina. In addition, some studies suggest a relationship between estimated connections and behavior. Xia et al. (2019) estimated the connection between the rat hippocampus (HPC) and prefrontal cortex (PFC). The results showed that the estimated connection corresponded to rat behavior, with strong connections between HPC and PFC observed only on trials where the rat responded correctly to the task.

3.2.2. Other models

In addition to the GLM or point-process models (Section 3.2.1), other models have been used to infer monosynaptic connections from spike trains. We introduce two popular models for connectivity inference, namely the ising model and the spiking neuron model.

- **Ising model:** Ising model is a simple mathematical model of a magnetic material in which each spin can take either an up or down state. A memory retrieval process in a network of model neurons is analogous to an energy relaxation process in a network of Ising spins. The Ising model can be used to analyze a network of neurons interacting through synapses. This can be done by transforming the spike trains into a binary state, resting or firing, with the time bin Δ. The connection between neurons can be estimated by fitting the model to a given set of spike data, such that each connection parameter represents the degree to which the firing activity of neurons is synchronized (Cocco et al., 2009; Posani et al., 2017). A problem with this model is that the estimation results vary greatly depending on the time bin Δ. To solve these problems, Terada et al. (2020) developed a kinetic Ising model that incorporates the effect of synaptic time delay and proposed a framework to optimize the bin width Δ.
- **Spiking neuron model:** The spiking neuron model describes how a neuron generates spikes in response to external input (Kobayashi et al., 2009; Gerstner et al., 2014). It has also been proposed to estimate the connections between neurons by fitting a network of spiking neuron models to the parallel spike data (Cocco et al., 2009; Isomura et al., 2015; Ladenbauer et al., 2019). It has also been proposed to estimate connections by training a spiking neural network with spike time-dependent plasticity (STDP) using measured spiking data (Moon et al., 2021).

3.3. Available source codes

As explained above, several techniques have been developed for estimating connections between neurons from spike data, and many of these techniques have been validated by applying them to simulated and experimental data. However, it often takes a lot of effort to implement these techniques programmatically. Therefore, it is convenient to use open source code. We list codes that are currently available (March 2024) in Table 2. Python and Matlab codes are becoming more popular. GLMCC and CoNNECT also have web applications that allow users to try analyzing spike data without programming. While TSPE, GLMCC, ExGLM are based on cross-correlation (Section 3.1.1), transfer entropy and CoTETE are based on information-theoretic measures (Section 3.1.2). Note that GLMCC and CoNNECT were validated by applying them to simulated data from the spiking neuron model and the Hodgkin-Huxley model.

4. Future challenges

We summarize the difficulties in estimating synaptic connections from spiking data and discuss possible future directions.

- **False positives:** While reasonable estimation methods provide many true positives for synaptic connections, there are cases where they provide false positives. The false positives can be caused by functional similarity or common inputs. The false estimation due to functional similarity can be eliminated by using the shift predictor (Fig. 3b: Perkel et al. (1967)), and the false estimation due to common inputs can be eliminated by the GLMCC (Fig. 3d: Kobayashi et al. (2019)) or the GLM (Kulkarni and Paninski, 2007). These methods incorporate the effect of the common inputs from other unobserved neurons in addition to the effect of synaptic connections between the recorded neurons. In addition, it important to note that the spike sorting algorithms are not perfect and may yield erroneous estimates of spike times, potentially due to experimental noise or double counting of spikes (Pillow et al., 2013). Therefore, it is essential to consider the possibility of such artifacts and incorporate them into the connectivity inference algorithms. Moreover, it would be interesting to utilize the cell type information that may be available from optogenetics experiments to improve the inference performance.
- **Competition of inference methods:** We have seen that different approaches for inferring synaptic connections have been proposed and validated using simulated data. It should be noted that the effective method may also depend on the simulation data. Since model developers tend to prefer data that have yielded favorable results, it is desirable that a third party organize a competition to infer synaptic connections from multicellular spike trains using experimental data in which the true connectivity is available for validation.
- **Network analysis:** Some studies have applied network analysis (Barabasi, 2018; Newman, 2018) to the estimated networks of neurons. Song et al. (2005) performed intracellular recordings from layer 5 pyramidal neurons in rat visual cortex and showed that the connections between neurons have non-random features, such that bidirectional connectivity and three-neuron connectivity patterns are observed more frequently than the random network (random graph). Furthermore, the strength of the connections follows a log-normal distribution, i.e. the synaptic connections consist of a few strong connections and many weak connections. Shimono and Beggs (2015) applied the transfer entropy method to spiking data measured with a 512-electrode array from cultured cells. They applied the network motif analysis (Milo et al., 2002) and the community detection algorithm (Blondel et al., 2008) to the inferred networks. The results showed that the connections are more tightly connected than in the random network, there are hub neurons connected to many cells, and large clusters (communities) connected by two or three synapses are found. The functional significance of the network structures in information processing needs to be investigated in the future.
- **Ca imaging data:** Recent advances in optical measurement techniques have been tremendous, particularly with respect to the number of neurons that can be recorded (Grienberger and Konnerth, 2012). While there have been some attempts to infer monosynaptic connections from Ca imaging data (Mishchencko et al., 2011; Stetter et al., 2012), it is still difficult to estimate synaptic connections from

Ca imaging data because the temporal resolution for spike time detection is longer than the typical synaptic delay of a few milliseconds. Most optical data have been used only to find "functional connections" or activity correlations between neurons (e.g., Ota et al. (2021)). Some technologies, such as the high-speed calcium imaging and voltage imaging (Bando et al., 2019; Kim and Schnitzer, 2022; Sakamoto and Yokoyama, 2024) are expected to overcome the time resolution limit. It is also worth mentioning that the first Neural Connectomics Challenge was organized with the goal of reconstructing monosynaptic connectivity from simulated Ca fluorescence signals (Guyon et al., 2014; de Abril et al., 2018). Furthermore, the influence of glial cells on neurons can be modeled (Nakae et al., 2014), as glial cell activity can be estimated from Ca imaging data. Such techniques may allow a more accurate estimation of synaptic connections between neurons by removing the influence of glial cells.

- **Closed-loop experimental design:** There is a growing interest in active, adaptive closed-loop (AACL) experiments (Chen and Pesaran, 2021) to overcome the curse of high-dimensional data, such as parallel spike trains or Ca imaging signal recorded from large number of neurons. Importantly, AACL experiments can design subsampling and resampling strategy of neurons guided by the experimenter's policy. In addition, AACL experiments can mitigate the causality problem via random external stimulus including optogenetic stimulation. Thus, it would be interesting to develop algorithms that improve the performance of detecting causal connectivity using AACL experiments.
- **Functional implications:** A future line of research could be to investigate the relationship between brain function and synaptic connectivity. Ko et al. (2011) measured synaptic connectivity in *vitro* and found that neighboring neurons with similar feature selectivity are preferentially connected. In addition, the methodology of inferring monosynaptic connectivity may allow us to study synaptic plasticity in behaving animals. Fujisawa et al. (2008) estimated synaptic connectivity from multiple spike trains recorded from the medial prefrontal cortex in rats and argued that the change in synaptic weight during the task could be explained by short-term facilitation and suppression. It is also important to examine the relationship between disease and synaptic connectivity. Experimental evidence suggests that schizophrenia might arise from reduced synaptic connectivity during brain development, including the perinatal and adolescent periods (McGlashan and Hoffman, 2000).
- **Improving the decoding accuracy:** It has been reported that information about interneuronal connectivity can be used to improve decoding performance. For example, the inferred connections can improve the reconstruction of visual information from retinal ganglion cell activity (Pillow et al., 2008) or the reconstruction of location information from hippocampal CA1 neuron activity (Posani et al., 2017). Evaluating decoding performance may be useful to investigate the functional significance of synaptic connectivity.

5. Conclusion

In this paper we have reviewed the methods for inferring or estimating monosynaptic connectivity from parallel spike trains. First, we have clarified a variety of meanings of "connectivity" or "connection" as the term is used differently in different areas of neuroscience research, such as structural connectivity, monosynaptic connectivity, and functional connectivity. Structural connectivity is defined as the anatomical connectivity between neurons or between brain regions. Anatomical connection is a sufficient condition, but it does not necessarily guarantee the presence of a monosynaptic connection. In contrast, functional connectivity simply means that there is a significant correlation of brain activity, which does not necessarily mean that there are synaptic connections between them.

Next, we focused on studies that infer monosynaptic connections from parallel spike trains of multiple neurons. As a result of efforts to refine the method, the accuracy of identifying monosynaptic connections seems to have improved dramatically in recent years. Nevertheless, it should be noted that the inference of synaptic connections will never be perfect (Brody, 1999; Gerstein et al., 1989; Stevenson et al., 2008b) or rule out the possibility of a connection based on the absence of a correlation (Stevenson, 2023). To confirm the existence of a monosynaptic connection, one must either stimulate the presynaptic neuron by juxtacellular/micro-stimulation or optogenetic stimulation (Swadlow, 1995; English et al., 2017), or manipulate the synaptic connections using synaptic blockers. However, due to the high cost of these experiments, it is impractical to do this for all neuron pairs. Therefore, a technique to estimate synaptic connections from spiking data would be useful to narrow down the number of neuron pairs to test.

CRediT authorship contribution statement

Ryota Kobayashi: Conceptualization, Formal analysis, Funding acquisition, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Shigeru Shinomoto:** Conceptualization, Supervision, Writing – review & editing.

Acknowledgments

This research was motivated by a special session at the annual Japanese neuroscience meeting held in Sendai, Japan on August 2023. We thank the speakers and participants of the special session and the financial support from 2023 JNS NSR/Elsevier Sponsored Symposium. This research was partially supported by JSPS KAKENHI (Nos. JP18K11560, JP19H01133, JP21H03559, JP21H04571, and JP22H03695) and AMED (Nos. JP21wm0525004 and JP223fa627001) to R.K, and JSPS KAKENHI (No. JP22H05163) to S.S.

References

de Abril, I.M., Yoshimoto, J., Doya, K., 2018. Connectivity inference from neural recording data: Challenges, mathematical bases and research directions. Neural Netw. 102, 120–137.
Aertsen, A., Gerstein, G., Habib, M., Palm, G., 1989. Dynamics of neuronal firing correlation: modulation of "effective connectivity. J. Neurophysiol. 61, 900–917.
Amarasingham, A., Harrison, M.T., Hatsopoulos, N.G., Geman, S., 2012. Conditional modeling and the jitter method of spike resampling. J. Neurophysiol. 107, 517–531.
Antonello, P.C., Varley, T.F., Beggs, J., Porcionatto, M., Sporns, O., Faber, J., 2022. Self-organization of in vitro neuronal assemblies drives to complex network topology. Elife 11, e74921.
Bando, Y., Sakamoto, M., Kim, S., Ayzenshtat, I., Yuste, R., 2019. Comparative evaluation of genetically encoded voltage indicators. Cell Rep. 26, 802–813.
Barabasi, A.L., 2018. Network Science. Oxford University Press.
Barthó, P., Hirase, H., Monconduit, L., Zugaro, M., Harris, K.D., Buzsáki, G., 2004. Characterization of neocortical principal cells and interneurons by network interactions and extracellular features. J. Neurophysiol. 92, 600–608.
Berry, M.J., Meister, M., 1998. Refractoriness and neural precision. J. Neurosci. 18, 2200–2211.
Binzegger, T., Douglas, R.J., Martin, K.A., 2004. A quantitative map of the circuit of cat primary visual cortex. J. Neurosci. 24, 8441–8453.
Blondel, V.D., Guillaume, J.L., Lambiotte, R., Lefebvre, E., 2008. Fast unfolding of communities in large networks. J. Stat. Mech. Theor. Exp. 2008, P10008.
Brillinger, D.R., 1975. The identification of point process systems. Ann. Probab. 909–924.
Brillinger, D.R., Bryant Jr, H.L., Segundo, J.P., 1976. Identification of synaptic interactions. Biol. Cybern. 22, 213–228.
Brody, C.D., 1999. Correlations without synchrony. Neural Comput. 11, 1537–1551.
Cai, Z., Neveu, C.L., Baxter, D.A., Byrne, J.H., Aazhang, B., 2017. Inferring neuronal network functional connectivity with directed information. J. Neurophysiol. 118, 1055–1069.
Chen, Z., Brown, E.N., 2022. Generalized linear models for point process analyses of neural spiking activity. Encyclopedia of Computational Neuroscience. Springer, pp. 1510–1513.
Chen, Z., Putrino, D.F., Ghosh, S., Barbieri, R., Brown, E.N., 2011. Statistical inference for assessing functional connectivity of neuronal ensembles with sparse spiking data. IEEE Trans. Neural Syst. Rehabil. Eng. 19, 121–135.
Chen, Z.S., Pesaran, B., 2021. Improving scalability in systems neuroscience. Neuron 109, 1776–1790.

Chornoboy, E., Schramm, L., Karr, A., 1988. Maximum likelihood identification of neural point process systems. Biol. Cybern. 59, 265–275.
Cocco, S. Leibler, S. Monasson, R. 2009.Neuronal couplings between retinal ganglion cells inferred by efficient inverse statistical physics methods.Proc. Natl. Acad. Sci.106, 14058-14062.
Cover, M., Thomas, J., 1991. Elements of information theory. John Wiley & Sons.
De Blasi, S., Ciba, M., Bahmer, A., Thielemann, C., 2019. Total spiking probability edges: A cross-correlation based method for effective connectivity estimation of cortical spiking neurons. J. Neurosci. Methods 312, 169–181.
van den Heuvel, M.P., Sporns, O., 2019. A cross-disorder connectome landscape of brain dysconnectivity. Nat. Rev. Neurosci. 20, 435–446.
Donner, C., Bartram, J., Hornauer, P., Kim, T., Roqueiro, D., Hierlemann, A., Obozinski, G., Schröter, M., 2024. Ensemble learning and ground-truth validation of synaptic connectivity inferred from spike trains. PLOS Comput. Biol. 20, e1011964.
Endo, D., Kobayashi, R., Bartolo, R., Averbeck, B.B., Sugase-Miyamoto, Y., Hayashi, K., Kawano, K., Richmond, B.J., Shinomoto, S., 2021. A convolutional neural network for estimating synaptic connectivity from spike trains. Sci. Rep. 11, 12087.
English, D.F., McKenzie, S., Evans, T., Kim, K., Yoon, E., Buzsáki, G., 2017. Pyramidal cell-interneuron circuit architecture and dynamics in hippocampal networks. Neuron 96, 505–520.
Ermentrout, B., 1998. Linearization of fi curves by adaptation. Neural Comput. 10, 1721–1729.
Evarts, E.V., 1968. Relation of pyramidal tract activity to force exerted during voluntary movement. J. Neurophysiol. 31, 14–27.
Friston, K.J., 1994. Functional and effective connectivity in neuroimaging: a synthesis. Hum. Brain Mapp. 2, 56–78.
Fujisawa, S., Amarasingham, A., Harrison, M.T., Buzsáki, G., 2008. Behavior-dependent short-term assembly dynamics in the medial prefrontal cortex. Nat. Neurosci. 11, 823.
Fujita, I., Tanaka, K., Ito, M., Cheng, K., 1992. Columns for visual features of objects in monkey inferotemporal cortex. Nature 360, 343–346.
Funahashi, S., Inoue, M., 2000. Neuronal interactions related to working memory processes in the primate prefrontal cortex revealed by cross-correlation analysis. Cereb. Cortex 10, 535–551.
Garofalo, M., Nieus, T., Massobrio, P., Martinoia, S., 2009. Evaluation of the performance of information theory-based methods and cross-correlation to estimate the functional connectivity in cortical networks. PloS ONE 4, e6482.
Gerhard, F., Kispersky, T., Gutierrez, G.J., Marder, E., Kramer, M., Eden, U., 2013. Successful reconstruction of a physiological circuit with known connectivity from spiking activity alone. PLoS Comput. Biol. 9, e1003138.
Gerstein, G.L., Perkel, D.H., 1969. Simultaneously recorded trains of action potentials: analysis and functional interpretation. Science 164, 828–830.
Gerstein, G.L., Bedenbaugh, P., Aertsen, A.M., 1989. Neuronal assemblies. IEEE Trans. Biomed. Eng. 36, 4–14.
Gerstner, W., Kistler, W.M., Naud, R., Paninski, L., 2014. Neuronal dynamics: From single neurons to networks and models of cognition. Cambridge University Press.
Gerwinn, S., Macke, J.H., Bethge, M., 2010. Bayesian inference for generalized linear models for spiking neurons. Front. Comput. Neurosci. 4, 1299.
Granger, C.W., 1969. Investigating causal relations by econometric models and cross-spectral methods. Econometrica 424–438.
Grienberger, C., Konnerth, A., 2012. Imaging calcium in neurons. Neuron 73, 862–885.
Guyon, I. Battaglia, D. Guyon, A. Lemaire, V. Orlandi, J.G. Ray, B. Saeed, M. Soriano, J. Statnikov, A. Stetter, O. 2014.Design of the first neuronal connectomics challenge: From imaging to connectivity, In: 2014 International Joint Conference on Neural Networks (IJCNN), IEEE.2600-2607.
Hawkes, A.G., 1971. Spectra of some self-exciting and mutually exciting point processes. Biometrika 58, 83–90.
Helmstaedter, M., Briggman, K.L., Turaga, S.C., Jain, V., Seung, H.S., Denk, W., 2013. Connectomic reconstruction of the inner plexiform layer in the mouse retina. Nature 500, 168–174.
Hubel, D.H., Wiesel, T.N., 1977. Ferrier lecture-functional architecture of macaque monkey visual cortex. Proc. R. Soc. Lond. Ser. B. Biol. Sci. 198, 1–59.
Isomura, T., Ogawa, Y., Kotani, K., Jimbo, Y., 2015. Accurate connection strength estimation based on variational bayes for detecting synaptic plasticity. Neural Comput. 27, 819–844.
Ito, S., Hansen, M.E., Heiland, R., Lumsdaine, A., Litke, A.M., Beggs, J.M., 2011. Extending transfer entropy improves identification of effective connectivity in a spiking cortical network model. PloS ONE 6, e27431.
Izhikevich, E.M., Edelman, G.M., 2008. Large-scale model of mammalian thalamocortical systems. Proc. Natl. Acad. Sci. USA 105, 3593–3598.
Jolivet, R., Rauch, A., Lüscher, H.R., Gerstner, W., 2006. Predicting spike timing of neocortical pyramidal neurons by simple threshold models. J. Comput. Neurosci. 21, 35–49.
Kim, S., Putrino, D., Ghosh, S., Brown, E.N., 2011. A granger causality measure for point process models of ensemble neural spiking activity. PLoS Comput. Biol. 7, e1001110.
Kim, T.H., Schnitzer, M.J., 2022. Fluorescence imaging of large-scale neural ensemble dynamics. Cell 185, 9–41.
Ko, H., Hofer, S.B., Pichler, B., Buchanan, K.A., Sjöström, P.J., Mrsic-Flogel, T.D., 2011. Functional specificity of local synaptic connections in neocortical networks. Nature 473, 87–91.
Kobayashi, R., 2009. The influence of firing mechanisms on gain modulation. J. Stat. Mech. Theor. Exp. 2009, P01017.
Kobayashi, R., Kitano, K., 2013. Impact of network topology on inference of synaptic connectivity from multi-neuronal spike data simulated by a large-scale cortical network model. J. Comput. Neurosci. 35, 109–124.
Kobayashi, R., Kitano, K., 2016. Impact of slow K^+ currents on spike generation can be described by an adaptive threshold model. J. Comput. Neurosci. 40, 347–362.
Kobayashi, R., Shinomoto, S., 2007. State space method for predicting the spike times of a neuron. Phys. Rev. E 75, 011925.
Kobayashi, R., Tsubo, Y., Shinomoto, S., 2009. Made-to-order spiking neuron model equipped with a multi-timescale adaptive threshold. Front. Comput. Neurosci. 3, 9.
Kobayashi, R., Kurita, S., Kurth, A., Kitano, K., Mizuseki, K., Diesmann, M., Richmond, B. J., Shinomoto, S., 2019. Reconstructing neuronal circuitry from parallel spike trains. Nat. Commun. 10, 4468.
Koch, C., 2004. Biophysics of computation: information processing in single neurons. Oxford University Press.
Kulkarni, J.E., Paninski, L., 2007. Common-input models for multiple neural spike-train data. Network 18, 375–407.
Kuroda, K., Ashizawa, T., Ikeguchi, T., 2011. Estimation of network structures only from spike sequences. Phys. A 390, 4002–4011.
Ladenbauer, J., McKenzie, S., English, D.F., Hagens, O., Ostojic, S., 2019. Inferring and validating mechanistic models of neural microcircuits based on spike-train data. Nat. Commun. 10, 4933.
Lambert, R.C., Tuleau-Malot, C., Bessaih, T., Rivoirard, V., Bouret, Y., Leresche, N., Reynaud-Bouret, P., 2018. Reconstructing the functional connectivity of multiple spike trains using hawkes models. J. Neurosci. Methods 297, 9–21.
Liew, Y.J., Pala, A., Whitmire, C.J., Stoy, W.A., Forest, C.R., Stanley, G.B., 2021. Inferring thalamocortical monosynaptic connectivity in vivo. J. Neurophysiol. 125, 2408–2431.
Liu, J.K., Schreyer, H.M., Onken, A., Rozenblit, F., Khani, M.H., Krishnamoorthy, V., Panzeri, S., Gollisch, T., 2017. Inference of neuronal functional circuitry with spike-triggered non-negative matrix factorization. Nat. Commun. 8, 149.
Lynall, M.E., Bassett, D.S., Kerwin, R., McKenna, P.J., Kitzbichler, M., Muller, U., Bullmore, E., 2010. Functional connectivity and brain networks in schizophrenia. J. Neurosci. 30, 9477–9487.
McGlashan, T.H., Hoffman, R.E., 2000. Schizophrenia as a disorder of developmentally reduced synaptic connectivity. Arch. Gen. Psychiatry 57, 637–648.
Milo, R., Shen-Orr, S., Itzkovitz, S., Kashtan, N., Chklovskii, D., Alon, U., 2002. Network motifs: simple building blocks of complex networks. Science 298, 824–827.
Mishchencko, Y., Vogelstein, J.T., Paninski, L., 2011. A bayesian approach for inferring neuronal connectivity from calcium fluorescent imaging data. Ann. Appl. Stat. 5, 1229–1261.
Mizuseki, K., Sirota, A., Pastalkova, E., Buzsáki, G., 2009. Theta oscillations provide temporal windows for local circuit computation in the entorhinal-hippocampal loop. Neuron 64, 267–280.
Mizuseki, K. Sirota, A. Pastalkova, E. Diba, K. Buzsáki, G. 2013.Multiple single unit recordings from different rat hippocampal and entorhinal regions while the animals were performing multiple behavioral tasks.CRCNS org.
Moon, J., Wu, Y., Zhu, X., Lu, W.D., 2021. Neural connectivity inference with spike-timing dependent plasticity network. Sci. China Inf. Sci. 64, 160405.
Moore, G.P., Segundo, J.P., Perkel, D.H., Levitan, H., 1970. Statistical signs of synaptic interaction in neurons. Biophys. J. 10, 876–900.
Nakae, K., Ikegaya, Y., Ishikawa, T., Oba, S., Urakubo, H., Koyama, M., Ishii, S., 2014. A statistical method of identifying interactions in neuron–glia systems based on functional multicell Ca^{2+} imaging. PLoS Comput. Biol. 10, e1003949.
Nedungadi, A.G., Rangarajan, G., Jain, N., Ding, M., 2009. Analyzing multiple spike trains with nonparametric granger causality. J. Comput. Neurosci. 27, 55–64.
Newman, M., 2018. Networks. Oxford University Press.
Nowak, L., Munk, M., James, A., Girard, P., Bullier, J., 1999. Cross-correlation study of the temporal interactions between areas V1 and V2 of the macaque monkey. J. Neurophysiol. 81, 1057–1074.
Okatan, M., Wilson, M.A., Brown, E.N., 2005. Analyzing functional connectivity using a network likelihood model of ensemble neural spiking activity. Neural Comput. 17, 1927–1961.
Ota, K., Oisi, Y., Suzuki, T., Ikeda, M., Ito, Y., Ito, T., Uwamori, H., Kobayashi, K., Kobayashi, M., Odagawa, M., et al., 2021. Fast, cell-resolution, contiguous-wide two-photon imaging to reveal functional network architectures across multi-modal cortical areas. Neuron 109, 1810–1824.
Palm, G., Aertsen, A., Gerstein, G., 1988. On the significance of correlations among neuronal spike trains. Biol. Cybern. 59, 1–11.
Paninski, L., 2004. Maximum likelihood estimation of cascade point-process neural encoding models. Network 15, 243.
Pastore, V.P., Massobrio, P., Godjoski, A., Martinoia, S., 2018. Identification of excitatory-inhibitory links and network topology in large-scale neuronal assemblies from multi-electrode recordings. PLoS Comput. Biol. 14, e1006381.
Perkel, D.H., Gerstein, G.L., Moore, G.P., 1967. Neuronal spike trains and stochastic point processes: II. simultaneous spike trains. Biophys. J. 7, 419–440.
Pernice, V., Rotter, S., 2013. Reconstruction of sparse connectivity in neural networks from spike train covariances. J. Stat. Mech. Theor. Exp. P03008.
Pillow, J.W., Shlens, J., Paninski, L., Sher, A., Litke, A.M., Chichilnisky, E., Simoncelli, E. P., 2008. Spatio-temporal correlations and visual signalling in a complete neuronal population. Nature 454, 995.
Pillow, J.W., Shlens, J., Chichilnisky, E., Simoncelli, E.P., 2013. A model-based spike sorting algorithm for removing correlation artifacts in multi-neuron recordings. PloS One 8, e62123.
Posani, L., Cocco, S., Ježek, K., Monasson, R., 2017. Functional connectivity models for decoding of spatial representations from hippocampal CA1 recordings. J. Comput. Neurosci. 43, 17–33.
Quinn, C.J., Coleman, T.P., Kiyavash, N., Hatsopoulos, N.G., 2011. Estimating the directed information to infer causal relationships in ensemble neural spike train recordings. J. Comput. Neurosci. 30, 17–44.

Reid, R.C., Alonso, J.M., 1995. Specificity of monosynaptic connections from thalamus to visual cortex. Nature 378, 281.
Ren, N., Ito, S., Hafizi, H., Beggs, J.M., Traveledson, I.H., 2020. Model-based detection of putative synaptic connections from spike recordings with latency and type constraints. J. Neurophysiol. 124, 1588–1604.
Sacerdote, L., Tamborrino, M., Zucca, C., 2012. Detecting dependencies between spike trains of pairs of neurons through copulas. Brain Res. 1434, 243–256.
Sakamoto, M. Yokoyama, T. Probing neuronal activity with genetically encoded calcium and voltage fluorescent indicators.Neurosci. Res. 2024, In press.
Scheffer, L.K., Xu, C.S., Januszewski, M., Lu, Z., Takemura, S.y., Hayworth, K.J., Huang, G.B., Shinomiya, K., Maitlin-Shepard, J., Berg, S., et al., 2020. A connectome and analysis of the adult drosophila central brain. Elife 9, e57443.
Schmidt, M., Bakker, R., Hilgetag, C.C., Diesmann, M., van Albada, S.J., 2018. Multi-scale account of the network structure of macaque visual cortex. Brain Struct. Funct. 223, 1409–1435.
Schreiber, T., 2000. Measuring information transfer. Phys. Rev. Lett. 85, 461.
Schwindel, C.D., Ali, K., McNaughton, B.L., Tatsuno, M., 2014. Long-term recordings improve the detection of weak excitatory–excitatory connections in rat prefrontal cortex. J. Neurosci. 34, 5454–5467.
Seth, A., 2007. Granger causality. Scholarpedia 2, 1667.
Shao, P.C., Huang, J.J., Shann, W.C., Yen, C.T., Tsai, M.L., Yen, C.C., 2015. Granger causality-based synaptic weights estimation for analyzing neuronal networks. J. Comput. Neurosci. 38, 483–497.
Shimono, M., Beggs, J.M., 2015. Functional clusters, hubs, and communities in the cortical microconnectome. Cereb. Cortex 25, 3743–3757.
Shorten, D.P., Spinney, R.E., Lizier, J.T., 2021. Estimating transfer entropy in continuous time between neural spike trains or other event-based data. PLoS Comput. Biol. 17, e1008054.
So, K., Koralek, A.C., Ganguly, K., Gastpar, M.C., Carmena, J.M., 2012. Assessing functional connectivity of neural ensembles using directed information. J. Neural Eng. 9, 026004.
Song, S., Sjöström, P.J., Reigl, M., Nelson, S., Chklovskii, D.B., 2005. Highly nonrandom features of synaptic connectivity in local cortical circuits. PLoS Biol. 3, e68.
Spivak, L., Levi, A., Sloin, H.E., Someck, S., Stark, E., 2022. Deconvolution improves the detection and quantification of spike transmission gain from spike trains. Commun. Biol. 5, 520.
Steinmetz, N.A., Aydin, C., Lebedeva, A., Okun, M., Pachitariu, M., Bauza, M., Beau, M., Bhagat, J., Böhm, C., Broux, M., et al., 2021. Neuropixels 2.0: A miniaturized high-density probe for stable, long-term brain recordings. Science 372, eabf4588.
Stetter, O., Battaglia, D., Soriano, J., Geisel, T., 2012. Model-free reconstruction of excitatory neuronal connectivity from calcium imaging signals. PLoS Comput. Biol. 8, e1002653.
Stevenson, I.H., 2023. Circumstantial evidence and explanatory models for synapses in large-scale spike recordings. Neurons Behav. Data Anal. Theory 1–30 (December).
Stevenson, I.H., Kording, K.P., 2011. How advances in neural recording affect data analysis. Nat. Neurosci. 14, 139.
Stevenson, I.H., Rebesco, J.M., Hatsopoulos, N.G., Haga, Z., Miller, L.E., Kording, K.P., 2008a. Bayesian inference of functional connectivity and network structure from spikes. IEEE Trans. Neural Syst. Rehabil. Eng. 17, 203–213.
Stevenson, I.H., Rebesco, J.M., Miller, L.E., Körding, K.P., 2008b. Inferring functional connections between neurons. Curr. Opin. Neurobiol. 18, 582–588.
Swadlow, H.A., 1995. Influence of vpm afferents on putative inhibitory interneurons in s1 of the awake rabbit: evidence from cross-correlation, microstimulation, and latencies to peripheral sensory stimulation. J. Neurophysiol. 73, 1584–1599.
Terada, Y., Obuchi, T., Isomura, T., Kabashima, Y., 2020. Inferring neuronal couplings from spiking data using a systematic procedure with a statistical criterion. Neural Comput. 32, 2187–2211.
Thomson, A.M., West, D.C., Wang, Y., Bannister, A.P., 2002. Synaptic connections and small circuits involving excitatory and inhibitory neurons in layers 2–5 of adult rat and cat neocortex: triple intracellular recordings and biocytin labelling in vitro. Cereb. Cortex 12, 936–953.
Toyama, K., Kimura, M., Tanaka, K., 1981. Organization of cat visual cortex as investigated by cross-correlation technique. J. Neurophysiol. 46, 202–214.
Truccolo, W., Eden, U.T., Fellows, M.R., Donoghue, J.P., Brown, E.N., 2005. A point process framework for relating neural spiking activity to spiking history, neural ensemble, and extrinsic covariate effects. J. Neurophysiol. 93, 1074–1089.
Tsubo, Y., Shinomoto, S., 2024. Nondifferentiable activity in the brain. PNAS Nexus 3 pgae261.
Vicente, R., Wibral, M., Lindner, M., Pipa, G., 2011. Transfer entropya model-free measure of effective connectivity for the neurosciences. J. Comput. Neurosci. 30, 45–67.
Xia, M., Liu, T., Bai, W., Zheng, X., Tian, X., 2019. Information transmission in hpc-pfc network for spatial working memory in rat. Behav. Brain Res. 356, 170–178.
Yoshimura, Y., Dantzker, J.L., Callaway, E.M., 2005. Excitatory cortical neurons form fine-scale functional networks. Nature 433, 868.
Yuan, M., Lin, Y., 2006. Model selection and estimation in regression with grouped variables. J. R. Stat. Soc. Ser. B: Stat. Method. 68, 49–67.
Zaytsev, Y.V., Morrison, A., Deger, M., 2015. Reconstruction of recurrent synaptic connectivity of thousands of neurons from simulated spiking activity. J. Comput. Neurosci. 39, 77–103.